

User Manual – Dragon Monster Game

Version: 1.0

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Language: Indonesian / English

Target Audience: Players, Developers, Testers

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Pengenalan Game

Apa itu Dragon Monster Game?

Dragon Monster Game adalah sebuah web-based RPG (Role-Playing Game) berbasis turn-based combat system. Dalam game ini, Anda berperan sebagai seorang petualang (hero) yang:

- Memilih hero dengan kemampuan unik
- Memilih pet dragon sebagai pendamping
- Bertarung dengan musuh dalam sistem pertempuran berbasis giliran
- Menjelajahi maps dengan level yang berbeda
- Menyimpan progress dalam save slots

Target Pemain

- Pemain casual yang menyukai RPG ringan
- Penggemar strategy-based combat
- Developer yang ingin belajar Spring Boot + JavaScript games

Fitur Utama

Hero Customization – Pilih atau buat hero dengan stat unik

Pet Dragon System – Pilih dragon companion dengan kemampuan berbeda

Multi-Level Maps – Jelajahi 3 peta dengan difficulty berbeda

Turn-Based Combat – Strategi pertempuran yang fair dan balanced

Experience & Leveling – Tingkatkan level dan stats hero Anda

Save System – Simpan hingga 10 game slot

Smooth Animations – Grafis HD dengan animation 60 FPS

Panduan Instalasi & Setup

Prasyarat Sistem

Sebelum bermain, pastikan sistem Anda memenuhi persyaratan:

Hardware Minimum

- RAM: 2 GB
- Storage: 500 MB
- Internet: Koneksi lokal atau internet (untuk multiplayer features - future)

Software yang Diperlukan

- Browser Modern: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- Java 25 (untuk menjalankan backend)
- PostgreSQL (opsional untuk production)

Langkah 1: Instal Java 25

Windows:

1. Download dari [oracle.com] (<https://www.oracle.com/java/technologies/downloads/>)
2. Jalankan installer `.exe`
3. Ikuti wizard instalasi

macOS:

```
```bash
brew tap homebrew/cask-versions
brew install java25
```
```

Linux (Ubuntu/Debian):

```
```bash
sudo apt update
sudo apt install openjdk-25-jdk
```
```

Langkah 2: Clone Repository

```
```bash
git clone https://github.com/MangBujang/Java-Game-Project.git
cd Java-Game-Project
```
```

Langkah 3: Jalankan Backend Server

macOS/Linux:

```
```bash
./mvnw spring-boot:run
```
```

Windows (PowerShell):

```
```powershell
.\mvnw.cmd spring-boot:run
```
```

Windows (Command Prompt):

```
```cmd
mvnw.cmd spring-boot:run
```
```

Output yang diharapkan:

```
```
Started DragonMonsterApplication in 5.123 seconds
```
```

Server berjalan di http://localhost:8081
```

Langkah 4: Buka Game di Browser

Ketika server sudah running, buka browser dan akses:

```

http://localhost:8081/game.html
```

Selesai! Game siap dimainkan. 🎉

Memulai Game

First Time Setup

Ketika Anda membuka game untuk pertama kali:

1. Loading Screen → Tunggu assets selesai dimuat (biasanya 2-3 detik)
2. Main Menu → Pilih "New Game" atau "Load Game"

Membuat Hero Baru

Jika memilih "New Game":

```

| SELECT YOUR HERO |
|---|
| 1. Wanderer Magician
- Health: 100
- Mana: 50
- Attack: 15
- Defense: 10
- Element: FIRE |
| 2. Knight Brave
- Health: 150
- Mana: 30
- Attack: 20
- Defense: 15
- Element: WATER |
| 3. Forest Ranger
- Health: 120
- Mana: 40
- Attack: 18
- Defense: 12
- Element: WIND |

Tekan angka untuk memilih hero, atau SPACE untuk lihat lebih banyak
```

Memilih Pet Dragon

Setelah memilih hero, pilih dragon companion:

...

SELECT YOUR PET
1. Fire Dragon (RARE) - Health: 80 - Attack: 25 - Ability: Flame Strike
2. Unicorn (RARE) - Health: 60 - Attack: 15 - Ability: Heal
3. Earth Guardian (EPIC) - Health: 100 - Attack: 20 - Ability: Stone Shield

Tekan angka untuk memilih pet

...

Memilih Difficulty

...

SELECT DIFFICULTY
EASY → Musuh lemah
NORMAL → Balanced
HARD → Musuh kuat
NIGHTMARE → Sangat sulit

...

Setelah memilih, tekan ENTER untuk mulai game!

Kontrol & Interface

Keyboard Controls

Tombol	Aksi	Konteks	
--	-----		
A atau ←	Left Arrow	Bergerak ke kiri	Gameplay
D atau →	Right Arrow	Bergerak ke kanan	Gameplay
W atau ↑	Up Arrow	Lompat	Gameplay
Space		Lompat (alternatif)	Gameplay
Mouse Click		Pilih menu	Menu/Dialog
Enter		Konfirmasi pilihan	Menu
Esc		Buka pause menu	Gameplay   Game

Interface

Top-Left: Player Stats

...

```
HERO: Wanderer Magician
Level: 1 | EXP: 0/100

HP: [██████████] 100/100
Mana: [██████░░░░] 50/50
Attack: 15 | Defense: 10
```

...

Bottom-Left: Pet Info

...

```
PET: Fire Dragon
HP: [██████░░░░] 80/80
Status: Following
```

...

Center: Game World

- Player character (dengan sprite animation)
- Pet dragon (mengikuti di belakang)
- Enemies (jika ada di map)
- Interactive objects (treasure, NPCs)

Bottom-Right: Minimap

...

```
Map 1/3
█ . . .
. . E .
. . . .
```

█ = Hero position

E = Enemy

. = Empty space

...

Top-Right: Quick Actions

...

[S] Save Game

[L] Load Game

[M] Map Select

[P] Pause

[I] Inventory (Future)

...

Game Mechanics

Movement & Navigation

Bergerak Horizontal

- Tekan A atau ← Left untuk bergerak ke kiri
- Tekan D atau → Right untuk bergerak ke kanan
- Hero akan terus bergerak selama tombol ditekan

### Lompat

- Tekan W atau Space untuk lompat
- Gravity effect membuat hero jatuh kembali
- Maximum jump height  $\approx$  150 pixel
- Dapat lompat sambil bergerak untuk kombinasi pergerakan

### Platform Collision

- Hero tidak bisa menembus ground/platform
- Ada gravity yang menarik hero ke bawah
- Jatuh dari platform membuat hero respawn di spawn point

### Parallax Scrolling

Map menggunakan parallax background untuk menciptakan depth effect:

...

```
[Sky] → 0% speed (stationary)
[Background Ruins] → 20% speed (slow)
[Hills & Trees] → 40% speed
[Ruins] → 60% speed
[Foreground] → 80% speed
[Ground] → 100% speed (fastest)
...
```

Saat Anda bergerak, layer yang lebih jauh bergerak lebih lambat, menciptakan efek 3D.

### Enemy Encounters

#### Deteksi Enemy

- Ketika player sprite menyentuh enemy sprite → combat dimulai
- Layar otomatis switch ke combat mode
- Music berubah ke combat theme

#### Multiple Enemies

- Map dapat memiliki multiple enemies spawn
- Enemy baru dapat spawn setelah defeat musuh sebelumnya
- Difficulty meningkat seiring progress

### Combat System

#### Combat Flow

...

1. ENCOUNTER ENEMY
└ Hero sprite touches enemy
2. COMBAT START
└ Hero stats displayed
└ Enemy stats displayed
└ Turn order determined (speed)
3. TURN PHASE
└ Hero selects action (Attack, etc)
└ Damage calculated
└ Enemy takes damage

Enemy AI takes action   Hero takes counter-damage
4. REPEAT until one defeated
5. COMBAT END    Victory → Get rewards   Defeat → Respawn or load save

...

## Combat Screen Layout

...

COMBAT INITIATED	
HERO: Wanderer Magician	ENEMY: Goblin
Level: 1 HP: [██████] 100/100	HP: [██████]
Attack: 15 Defense: 10	Attack: 10
Element: FIRE	Element: EARTH
Your Turn! Select Action:	
[1] ATTACK	- Deal normal damage
[2] SPECIAL	- Use element advantage (5 MP)
[3] DEFEND	- Reduce incoming damage
[4] ITEM	- Use consumable (future)
► Your choice: _	

...

## Combat Actions

### 1. Attack (Normal)

- Cost: None
- Effect: Deal  $(\text{Attack} + \text{random}(1-10)) - (\text{Enemy Defense} / 2)$  damage
- Description: Serangan normal tanpa efek khusus

Contoh:

...

Your Attack:  $15 + 5 \text{ (random)} = 20$   
Enemy Defense reduction:  $10 / 2 = 5$   
Final Damage:  $20 - 5 = 15$  damage  
Enemy HP: 40 → 25

...

### 2. Special Attack

- Cost: 10 Mana (MP)
- Effect: Damage  $\times 1.5$  jika element advantage
- Description: Serangan khusus dengan bonus element

Element Advantage:

- FIRE beats EARTH → 1.5x damage
- WATER beats FIRE → 1.5x damage
- EARTH beats WATER → 1.5x damage
- WIND beats WATER → 1.5x damage

Contoh:

```

Fire Hero vs Earth Enemy:

Normal Attack: 15 damage

Special Attack: $15 \times 1.5 = 22$ damage (using 10 MP)

```

### 3. Defend

- Cost: None

- Effect: Reduce incoming damage by 50%

- Description: Posisi defensive untuk mengurangi damage

Contoh:

```

Normal Enemy Attack: 10 damage

While Defending: $10 \times 0.5 = 5$ damage (reduced)

```

### 4. Item (Future)

- Planned feature untuk consumable items

- Currently disabled

### Damage Calculation Formula

```

Base Damage = Attacker Attack Power + Random(1-10)

Defense Reduction = Defender Defense / 2

Element Multiplier = Check Element Advantage (1.0x or 1.5x)

Final Damage = (Base Damage - Defense Reduction) × Element Multiplier

Minimum Damage = 1 (never 0 or negative)

Maximum Damage = Unlimited (based on stats)

```

### Example Combat Scenario

```

Turn 1:

Hero (Wanderer Magician) vs Goblin

Hero: "Attack!"

→ Damage: 20

→ Goblin HP: 40 → 20

Goblin: "Attack!"

→ Damage: 8 (counter-attack)

→ Hero HP: 100 → 92

Turn 2:

Hero: "Special Attack!" (FIRE vs EARTH)

→ Damage: 30 (with 1.5x element advantage)

→ Goblin HP: 20 → 0 (DEFEATED!)

Hero wins!

Rewards:

+ 30 Experience Points

+ 15 Gold

+ Health Potion (item)

Hero EXP: 0 → 30
Next Level: 30/100
...

Victory & Defeat

Victory Screen
...

| VICTORY! |
|--|
| You defeated Goblin Warrior |
| Rewards:
+ 30 Experience Points
+ 15 Gold
+ Health Potion |
| Hero EXP: 0 → 30
Level: 1/2 (30/100 XP to level up) |
| [CONTINUE] [SAVE & CONTINUE] |

...

Defeat Screen

...

| YOU LOST! |
|---|
| You were defeated by Goblin Warrior |
| Consolation:
+ 5 Experience Points (50% penalty)
+ 0 Gold |
| Hero EXP: 0 → 5 |
| [RETRY] [LOAD SAVE] [MAIN MENU] |

...

Experience & Leveling

Experience System

Gaining Experience

- Victory: Full reward (50 XP base)
- Defeat: 50% penalty (25 XP)
- XP varies by enemy level and difficulty

Leveling Progression

...

Level 1 → Level 2: Requires 100 XP
Level 2 → Level 3: Requires 200 XP
Level 3 → Level 4: Requires 300 XP

Level $n \rightarrow$ Level $n+1$: Requires $n \times 100$ XP

Formula: Required XP = Current Level $\times$ 100
```

Stat Increases per Level  
```

On Level Up:
+ 10 Health Points (HP)
+ 2 Attack Power
+ 1 Defense
+ Mana regenerates to max
```

Level Cap  
- Maximum level: 100  
- After level 100, EXP still counts but no further stat growth

Level-Up Example

```

Status: Level 1
Current EXP: 80 / 100 XP

After defeating 3 enemies (50 XP each):
Current EXP: $80 + 50 + 50 + 50 = 230$ XP

Level Up! 🎉
Level 1 $\rightarrow$ Level 2
EXP carries over: $230 - 100 = 130$ / 200 XP

Stats Updated:
HP: 100 $\rightarrow$ 110
Attack: 15 $\rightarrow$ 17
Defense: 10 $\rightarrow$ 11
```

Manajemen Save/Load

Save System Overview

Anda dapat menyimpan progress di hingga 10 save slots. Cara

Menyimpan Game

Method 1: In-Game Save

1. Tekan S atau pilih [Save Game] di menu
2. Pilih save slot (1-10) atau buat slot baru
3. Masukkan nama save (misal: "Progress Map 1")
4. Tekan SAVE untuk confirm

Method 2: Quick Save

1. Tekan Ctrl+S (quick save)
2. Otomatis menyimpan ke slot terbaru

Cara Memuat Game

Method 1: Main Menu

1. Di main menu, pilih [Load Game]
2. Pilih save yang ingin dimuat
3. Tekan LOAD

Method 2: In-Game Load

1. Tekan L atau pilih [Load Game]
2. Pilih save yang ingin dimuat
3. Konfirmasi

Save Slot Information

...

SAVE SLOTS
[1] Progress Map 1 Level: 5   EXP: 45/500   Gold: 150 Hero: Wanderer Magician   Pet: Fire Drg Created: 2024-07-23 10:30 Last Saved: 2024-07-23 12:45
[2] Boss Defeated! Level: 12   EXP: 230/1200   Gold: 500 Hero: Knight Brave   Pet: Unicorn Created: 2024-07-22 18:00 Last Saved: 2024-07-22 23:30
[3] Empty Slot Click to create new save

Navigation: ← → to select | ENTER to load | DELETE to remove

...

What Gets Saved Save

slot menyimpan:

- Hero stats (level, health, experience)
- Pet/dragon selection
- Current position on map
- Inventory (items/gold)
- Map progression
- Combat state
- Chat history (tidak tersimpan)
- Settings (tidak tersimpan per save)

Maps Overview

Map 1: Ruins

Difficulty: Easy

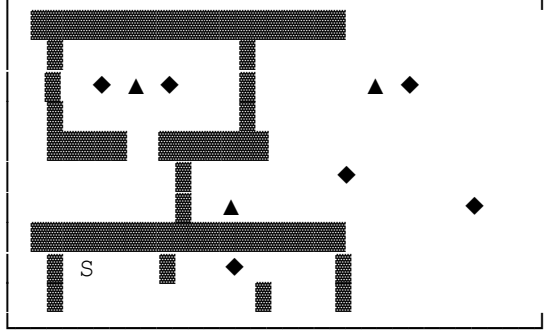
Location: Ancient Ruins of Fire

Theme: Desert with ancient structures

Enemy Type: Goblins, Fire Elementals

...

## MAP 1: RUINS



S = Start Point

▲ = Hero

◆ = Enemy

■ = Platform

\\

### Objectives:

- Defeat 3 enemies
- Reach map end
- Unlock Map 2

## Map 2: Forest

Difficulty: Normal

Location: Ancient Forest

Theme: Forest with trees and water

Enemy Type: Forest Beasts, Water Spirits

### Objectives:

- Defeat 5 enemies
- Find treasure chest
- Unlock Map 3

## Map 3: Castle

Difficulty: Hard

Location: Dragon's Castle

Theme: Dark castle with flying enemies

Enemy Type: Knights, Dragons, Dark Sorcerers

### Objectives:

- Defeat boss (Dragon)
- Complete final challenge
- Unlock endgame content

## Tips & Trik

### Combat Strategy

#### Tip 1: Manfaatkan Element Advantage

- Ketahui elemen lawan sebelum bertarung
- Gunakan special attack jika ada advantage
- Element advantage memberikan 1.5x damage!

#### Tip 2: Manajemen Mana

\\

Mana = Resource berharga

- Gunakan special attack hanya saat diperlukan
  - Kumpulkan mana dengan tidak bertarung
  - Mana regenerates fully setelah combat
- ...

Tip 3: Pertahankan HP

- Jangan berhemat dengan defend action
- Defend mengurangi damage 50%
- Lebih baik 1 turn defend dari 2 turn damage

Tip 4: Pilih Pet yang Tepat

- Fire Dragon: Damage output tinggi
- Unicorn: Support & healing
- Earth Guardian: Tank & defense
- Tip: Sesuaikan dengan playstyle Anda

## Progression Tips

Leveling Cepat

...

1. Pilih difficulty HARD
  2. Setiap victory memberikan 50+ XP
  3. Defeat enemies di level terakhir yang sudah dikuasai
  4. Farm XP selama 10-15 menit per level
- ...

Save Management

...

- Save sebelum boss fight
  - Jangan overwrite save slot penting
  - Rotate antara 3-4 save slots
  - Backup save slot setiap achievement
- ...

Gold Farming

...

- Enemies memberikan gold setiap defeat
  - Difficulty HARD → More gold
  - Level map terakhir → Most gold
  - Farming 30 min = 500+ gold
- ...

## Quality of Life Tips

1. Gunakan Keyboard Shortcuts

- S = Save
- L = Load
- Esc = Pause menu
- M = Map select

2. Optimize Graphics

- Disable background animations jika lag
- Browser fullscreen untuk immersion
- Brightness +10% untuk visibility

3. Audio Settings

- Mute background music jika playing sambil konsentrasi
- Sound effects untuk feedback

- Volume: 50-70% ideal

## Troubleshooting

### Issue 1: Game Tidak Bisa Diakses

Error: "Cannot connect to localhost:8081"

Solusi:

```bash

Check if server running

ps aux | grep java

If not running, start server

./mvnw spring-boot:run

Wait for "Started DragonMonsterApplication" message

Then try http://localhost:8081 again

```

### Issue 2: Assets Tidak Loading

Error: "Failed to load image: assets/..."

Solusi:

1. Check browser console (F12 → Console tab)

2. Verify assets folder exists at `src/main/resources/static/assets/`

3. Clear browser cache (Ctrl+Shift+Delete)

4. Reload page (Ctrl+Shift+R)

### Issue 3: Combat Damage Tidak Terhitung

Error: "Damage shows 0"

Solusi:

1. Verify hero attack stat > 0

2. Enemy defense tidak melebihi hero attack

3. Minimum damage selalu 1

4. Check browser console untuk error API

### Issue 4: Save/Load Tidak Bekerja

Error: "Save failed" atau "Load failed"

Solusi:

```bash

Check database connection

In application.properties, verify:

spring.datasource.url=jdbc:postgresql://localhost:5432/dragon_monster

spring.datasource.username=game_user

spring.datasource.password=correct_password

```
Restart server if database changed
./mvnw spring-boot:run
```
```

#### Issue 5: Game Lag atau FPS Rendah

Symptoms: Stutter, choppy animation, slow response

Solusi:

1. Close other browser tabs
2. Disable browser extensions
3. Lower graphics quality (if available)
4. Restart browser
5. Check RAM usage (Task Manager / Activity Monitor)

#### Issue 6: Keyboard Controls Tidak Responsive

Error: Hero tidak bergerak meski tombol ditekan

Solusi:

1. Click pada game canvas dulu (focus element)
2. Check NUM LOCK status
3. Try mouse controls (if available)
4. Restart browser

#### Issue 7: Browser Compatibility

Error: "Game crashes" atau "Features not working"

Solusi:

- Gunakan browser modern yang didukung:
  - Chrome 90+
  - Firefox 88+
  - Safari 14+
  - Edge 90+
- Hindari: IE11, old Safari versions

#### FAQ (Frequently Asked Questions)

Q1: Berapa lama game ini?

A: Tergantung playstyle:

- Casual: 3-5 jam untuk complete all maps
- Speedrun: 1-2 jam
- 100% with all achievements: 10+ jam

Q2: Berapa max level yang bisa dicapai?

A: Maximum level adalah 100. Setelah mencapai level 100:

- Stat tidak bertambah lagi
- Experience masih bisa dikumpulkan (untuk leaderboard future)
- Bonus achievement unlocked

Q3: Apakah ada multiplayer?

A: Saat ini, game ini single-player only. Fitur multiplayer direncanakan untuk versi future.

Q4: Berapa banyak save slots?

A: Anda dapat membuat hingga 10 save slots. Setelah penuh, Anda perlu delete slot lama untuk membuat yang baru.

Q5: Apakah save bisa corrupt?

A: Save slot disimpan di database PostgreSQL dan jarang corrupt. Tapi untuk safety:

- Backup important saves
- Jangan force-close browser saat saving
- Jangan ubah database files manual

Q6: Bisakah saya reset character?

A: Ya, dengan cara:

1. Delete save slot yang ingin direset
2. Buat new game

Q7: Apa elemen yang paling kuat?

A: Tidak ada elemen "paling kuat" karena balance:

- FIRE → beats EARTH (DPS tinggi)
- WATER → beats FIRE (Balanced)
- EARTH → beats WATER (Tank)
- WIND → beats WATER (Speed)

Pilih berdasarkan playstyle!

Q8: Bagaimana cara mendapat banyak gold?

A: Tips farming gold:

1. Play on HARD difficulty
2. Defeat high-level enemies
3. Complete map 3 repeatedly
4. Average: 50-100 gold per enemy



Q9: Apakah ada hidden content?

A: Tidak ada hidden content di versi 1.0. Tapi stay tuned untuk:

- Secret boss battles
- Legendary items
- Unlock able area
- Easter eggs

Q10: Support & Bug Report?

A: Untuk report bug atau suggest feature:

1. Buka GitHub issue:  
[MangBujang/Java-Game-Project/issues] (<https://github.com/MangBujang/Java-Game-Project/issues>)
2. Deskripsikan bug dengan detail
3. Include screenshot/video jika possible
4. Include system info (OS, browser, Java version)

Kontak & Support

Untuk Pemain

- Bug Report: [GitHub Issues] (<https://github.com/MangBujang/Java-Game-Project/issues>)
- Discussion: [GitHub Discussions] (<https://github.com/MangBujang/Java-Game-Project/discussions>)
- Email: roffil404@gmail.com

Untuk Developer

- Installation Guide: [INSTALLATION\_GUIDE.md] (./INSTALLATION\_GUIDE.md)
- Functional Requirements: [FUNCTIONAL\_REQUIREMENTS.md] (./FUNCTIONAL\_REQUIREMENTS.md)
- Documentation: [README.md] (./README.md)

Learning Resources

Untuk Belajar Spring Boot

- [Spring Boot Official Documentation] (<https://spring.io/projects/spring-boot>)
- [Spring Data JPA Guide] (<https://spring.io/guides/gs/accessing-data-jpa/>)
- [RESTful Web Services] (<https://spring.io/guides/gs/rest-service/>)

Untuk Belajar JavaScript Gaming

- [HTML5 Canvas API] ([https://developer.mozilla.org/en-US/docs/Web/API/Canvas\\_API](https://developer.mozilla.org/en-US/docs/Web/API/Canvas_API))
- [RequestAnimationFrame] (<https://developer.mozilla.org/en-US/docs/Web/API/window/requestAnimationFrame>)
- [Game Development Basics] ([https://developer.mozilla.org/en-US/docs/Games/Tutorials/2D\\_breakout\\_game\\_Phaser](https://developer.mozilla.org/en-US/docs/Games/Tutorials/2D_breakout_game_Phaser))

Untuk Belajar Game Design

- [Game Mechanics 101] (<https://www.gamasutra.com/>)
- [Balance in Game Design] (<https://www.youtube.com/watch?v=e31SCULXPU0>)

- [Combat System Design] (<https://www.youtube.com/watch?v=c38qBCgXqTM>)

#### Checklist First Time Player

- [ ] Server sudah running di localhost:8081
- [ ] Browser terbuka di <http://localhost:8081/game.html>
- [ ] Assets sudah loading (tidak ada red X di console)
- [ ] Pilih hero untuk dimulai
- [ ] Pilih pet dragon
- [ ] Coba combat pertama
- [ ] Save game pertama
- [ ] Jelajahi map 2 & 3
- [ ] Capai level 5+
- [ ] Defeat boss akhir
- [ ] Baca FAQ jika ada pertanyaan

Selamat Bermain!

Anda sudah siap menjadi ultimate dragon monster master!

Tips akhir:

- Nikmati gameplay dan jangan terburu-buru
- Experiment dengan hero dan pet berbeda
- Challenge diri dengan HARD difficulty
- Share progress Anda dengan teman!

Terima kasih sudah memainkan Dragon Monster Game!

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License: Open Source